

The primary structure of a protein is a series of amino acids that are connected in a long chain via C–N peptide bonds (Figure 1).

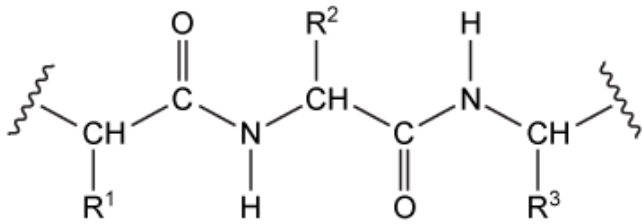


Figure 1 Peptide bonds within a protein segment (Note: R^n = amino acid side chains)

Protein chain folding is guided by multiple intramolecular interactions that can occur between two peptide bonds, between a peptide bond and a side chain, or between two side chains, as in the protein shown in Figure 2. Although these interactions are intramolecular because they occur within the same molecule, they act effectively like intermolecular interactions because proteins are such large molecules.

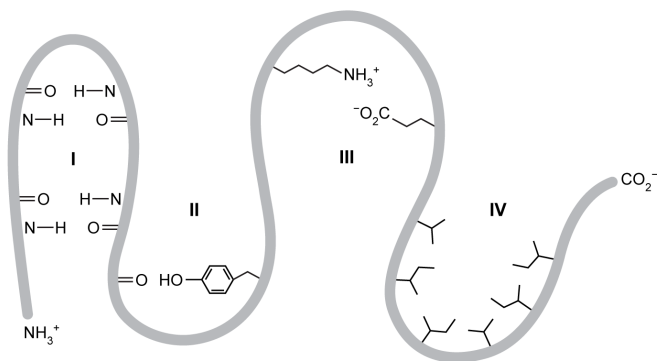


Figure 2 Intramolecular interactions between the peptide chain and amino acid side chains within a hypothetical protein

Intramolecular interactions play a key role in the folding of a protein into a three-dimensional structure. As such, the water solubility of a folded protein depends largely on the properties of the side chains exposed to the outer surface of the folded protein structure.

Which of the following regions from the protein shown in Figure 2 depict hydrogen bonding and a salt bridge, respectively?

- ☐ A. I and II [11%]

✔ ☒ B. II and III [85%]

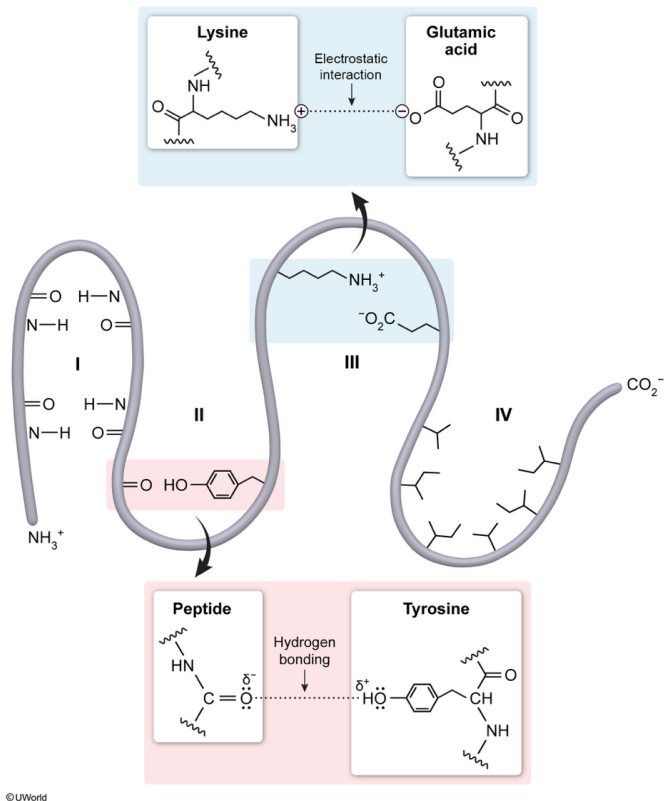
☐ C. II and IV [1%]

☐ D. III and IV [2%]

Correct

84% answered correctly

Explanation:



Hydrogen bonding is a special type of **dipole-dipole interaction** in which at least one of the **dipoles** involves a hydrogen atom that is **covalently bonded** to an atom of oxygen, nitrogen, or fluorine. In the protein shown in Figure 2, regions I and II both depict attractions associated with hydrogen bonding. Region I has hydrogen bonding between the N-H bonds (ie, the hydrogen bond **donors**) and the C=O bonds (ie, the hydrogen bond acceptors) from the peptide backbone. Region II has a hydrogen bond between the oxygen atom of a C=O bond in the peptide chain and the O-H bond in the tyrosine side chain.

When two **oppositely charged side chains** are close enough to each other, they experience a strong, mutual **electrostatic attraction** and form an **ion-ion interaction** known as a **salt bridge**. The only depiction of a salt bridge in Figure 2 is in region III, where the side chain of **lysine** ($-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_3^+$) interacts with the side chain of **glutamic acid**

($-\text{CH}_2\text{CH}_2\text{COO}^-$).

Therefore, regions II and III depict hydrogen bonding and a salt bridge, respectively.

(Choice A) Regions I and II *both* involve hydrogen bonding. Neither has a charged species that can participate in an electrostatic interaction necessary to form a salt bridge.

(Choice C) Region II does have hydrogen bonding, but region IV contains only **London dispersion forces** between **nonpolar** amino acid side chains (ie, no ionic bonding).

(Choice D) Region III contains an electrostatic ion-ion interaction (ie, a salt bridge), and region IV involves only London dispersion interactions.

Educational objective:

Hydrogen bonding is a special type of dipole-dipole interaction in which at least one of the dipoles involves a hydrogen atom covalently bonded to one of the highly electronegative elements oxygen, nitrogen, or fluorine. Salt bridges form from strong electrostatic attractions (eg, ion-ion interactions) between two oppositely charged side chains in a protein.

Time spent:0 sec

Subject:
General Chemistry

QID:401770

System:
Interactions of Chemical Substances

The primary structure of a protein is a series of amino acids that are connected in a long chain via C–N peptide bonds (Figure 1).

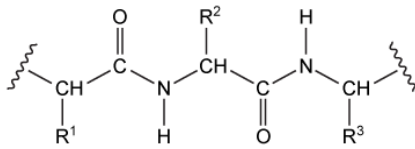


Figure 1 Peptide bonds within a protein segment (Note: R^n = amino acid side chains)

Protein chain folding is guided by multiple intramolecular interactions that can occur between two peptide bonds, between a peptide bond and a side chain, or between two side chains, as in the protein shown in Figure 2. Although these interactions are intramolecular because they occur within the same molecule, they act effectively like intermolecular interactions because proteins are such large molecules.

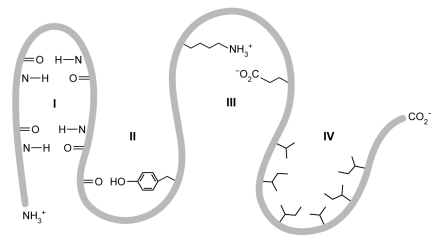


Figure 2 Intramolecular interactions between the peptide chain and amino acid side chains within a hypothetical protein

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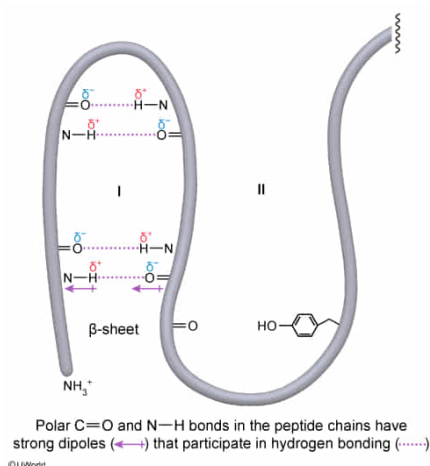
In the protein shown in Figure 2, the folding of region I is the result of:

- ☐ A. salt bridge interactions between amino acid side chains.
- ☐ B. covalent bonding between peptide segments. [5%]
- ☐ C. hydrophobic interactions between amino acid side chains. [5%]
- ☒ D. hydrogen bonding between functional groups that make up the backbone of the peptide chain. [87%]

Correct

88% answered correctly

Explanation:



Hydrogen bonding is a special type of dipole-dipole interaction in which at least one of the dipoles involves a hydrogen

微信Mai_ange

atom that is covalently bonded to an atom of oxygen, nitrogen, or fluorine. In such a bond, the large difference in electronegativity between the atoms causes an unequal sharing of electrons that gives the hydrogen atom a partial positive charge (δ^+) and the electronegative atom a partial negative charge (δ^-). Attractions between these partial charges result in hydrogen bond donor and acceptor pairs that form strong noncovalent interactions (ie, intermolecular forces).

Hydrogen bonding is one of the strongest intermolecular (and noncovalent intramolecular) forces, considerably stronger than other types of dipole-dipole attractions. This strength is due to the large dipole moments involved as well as the small size of the hydrogen atom. This small size allows the partially positive hydrogen to be closer to a partially negative atom, which creates a stronger attraction.

In region I of the protein shown in Figure 2, the partial positive charge from the dipole of the N–H bond (amino group) is attracted to the partial negative charge from the dipole of the C=O bond (carboxyl group). This type of hydrogen bonding interaction that forms between the functional groups that are part of backbone of the peptide chain result in the formation of a β -sheet secondary structure in the protein.

(Choice A) A salt bridge interaction is a linkage formed by the electrostatic attractions between two fully charged amino acid side chains. This connection contrasts with species attracted because of a dipole moment involving *partial* charges induced by electronegativity differences.

(Choice B) The interactions in region I are noncovalent because the atoms are not bound by shared electrons.

(Choice C) Hydrophobic interactions occur between molecules that are nonpolar (ie, lack a significant permanent dipole) and have very low water solubility. The dipole interactions of region I are not hydrophobic.

Educational objective:

Hydrogen bonding is one of the strongest intermolecular (and noncovalent intramolecular) forces. It is a special kind of noncovalent dipole-dipole interaction in which one of the dipoles involves a hydrogen atom covalently bonded to a highly electronegative atom of nitrogen, oxygen, or fluorine.

Time spent:0 sec
Subject:
General Chemistry

QID:401771
System:
Interactions of Chemical Substances

The primary structure of a protein is a series of amino acids that are connected in a long chain via C–N peptide bonds (Figure 1).

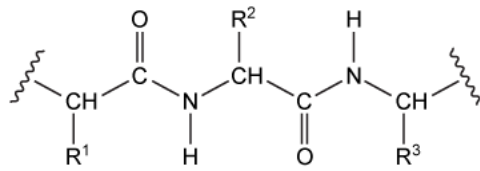


Figure 1 Peptide bonds within a protein segment (Note: R^n = amino acid side chains)

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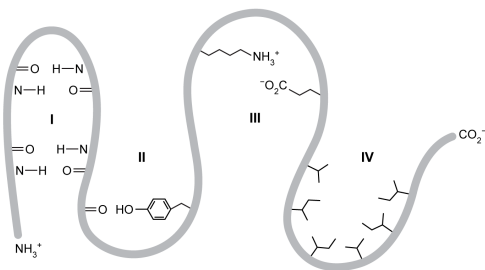


Figure 2 Intramolecular interactions between the peptide chain and amino acid side chains within a hypothetical protein

Intramolecular interactions play a key role in the folding of a protein into a three-dimensional structure. As such, the water solubility of a folded protein depends largely on the properties of the side chains exposed to the outer surface of the folded protein structure.

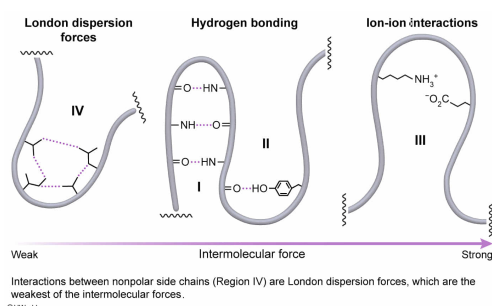
What region of the protein shown in Figure 2 depicts the weakest type of interaction?

- ☐ A. I
- ☐ B. II [1%]
- ☐ C. III [2%]
- ☒ D. IV [95%]

Correct

95% answered correctly

Explanation:



Covalent bonds are strong chemical bonds formed when two atoms share electrons. In contrast, **noncovalent interactions** are weak attractions between atoms that do not share electrons but have dipoles with opposite partial charges. Noncovalent dipole interactions include attractions between two permanent dipoles, attractions between a permanent dipole and an induced dipole, and attractions between two induced dipoles. These attractions are collectively called **van der Waals forces**.

London dispersion forces are among the weakest of the van der Waals forces (as an individual interaction). However, the combined effect of numerous dispersion force interactions often make London dispersion forces a very strong net influence on physical properties, especially in larger molecules. These interactions occur between two molecules whose proximity to each other induces a mutual attraction via the creation of a weak, temporary **dipole moment**.

Within a nonpolar molecule, the average electron distribution does not concentrate around any particular nucleus sufficiently enough to produce a permanent dipole moment. However, when two nonpolar molecules come near each other, the electron cloud of one molecule can momentarily shift and form a localized, slightly positive region that then induces a slightly negative region in a neighboring molecule. Consequently, **instantaneous dipoles** with weak mutual attraction are momentarily formed.

In region IV of the protein shown in Figure 2, only nonpolar alkyl side chains (ie, no permanent dipoles) are found. Therefore, the primary interactions in region IV are London dispersion forces (the weakest type).

(Choices A and B) Regions I and II of the protein both involve **hydrogen bonding**, which is one of the stronger noncovalent interactions, not the weakest. In regions I and II, the oxygen from the C=O bond interacts with the hydrogen dipole from either an N–H or an O–H bond.

(Choice C) In region III of the protein, a salt bridge is formed between two charged atoms in the side chains. This is an ion-ion interaction, which is the *strongest* noncovalent interaction represented in this protein.

Educational objective:

London dispersion forces are the weakest of the noncovalent van der Waals forces. These interactions occur between two atoms or molecules near enough to each other for distortions in the electron cloud to induce weak instantaneous dipoles.

Time spent:0 sec

Subject:
General Chemistry

QID:401772

System:
Interactions of Chemical Substances

The primary structure of a protein is a series of amino acids that are connected in a long chain via C–N peptide bonds (Figure 1).

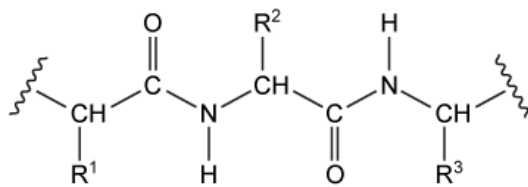


Figure 1 Peptide bonds within a protein segment (Note: R^n = amino acid side chains)

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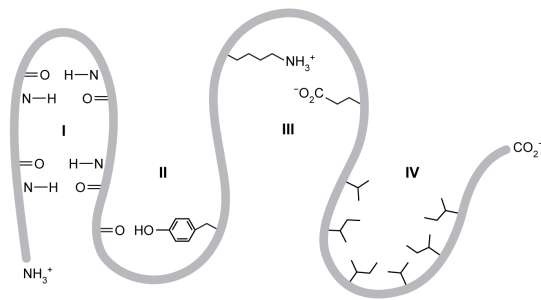


Figure 2 Intramolecular interactions between the peptide chain and amino acid side chains within a hypothetical protein

Intramolecular interactions play a key role in the folding of a protein into a three-dimensional structure. As such, the water solubility of a folded protein depends largely on the properties of the side chains exposed to the outer surface of the folded protein structure.

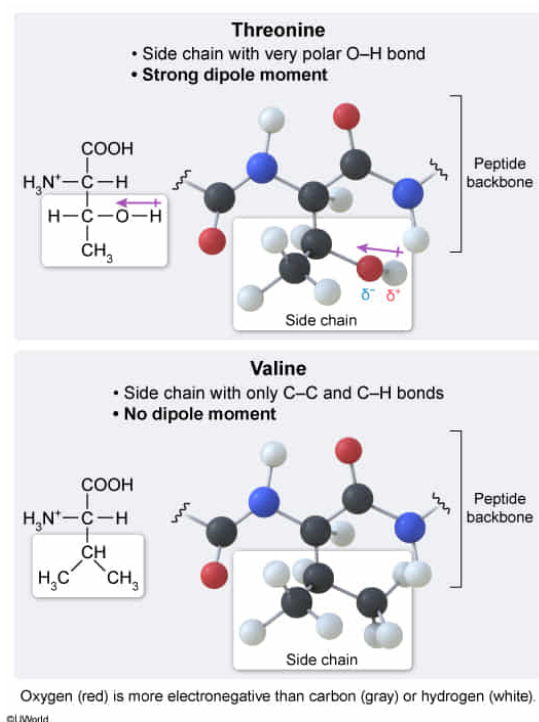
Which statement about the side chains of threonine and valine is true?

- ☐ A. Both threonine and valine have side chains with strong dipoles that can participate in dipole-dipole interactions. [3%]
- ✓ ☒ B. Threonine has a side chain with a strong dipole that can participate in dipole-dipole interactions but valine does not. [86%]
- ☐ C. Valine has a side chain with a strong dipole that can participate in dipole-dipole interactions but threonine does not. [3%]
- ☐ D. Neither threonine nor valine has a side chain with a strong dipole that can participate in dipole-dipole interactions. [5%]

Correct

86% answered correctly

Explanation:



When two atoms with a large difference in **electronegativity** are covalently bonded together, the more electronegative atom draws electron density away from the less electronegative atom and creates a **polar bond** with a permanent **dipole moment**. When two **dipoles** from different molecules are near each other, **opposite partial charges** within the dipoles experience a **mutual attraction** and align such that the partially positive (δ^+) portion of one dipole interacts with the partially negative (δ^-) portion of the other dipole. This is known as a **dipole-dipole interaction**.

The side chain of threonine contains a highly polar O–H bond, which gives threonine a strong, permanent dipole that can participate in dipole-dipole interactions. Conversely, the side chain of valine contains only nonpolar hydrocarbon bonds that do not have strong, permanent dipoles to participate in dipole-dipole interactions.

(Choices A, C, and D) Although threonine has a **polar side chain** with a highly polar O–H bond that can participate in dipole-dipole interactions, valine has a **hydrophobic side chain** with only nonpolar C–C and C–H bonds, which do not have a strong, permanent dipole to participate in dipole-dipole interactions.

Educational objective:

Dipole-dipole interactions are a type of noncovalent interaction that occurs between neighboring polar molecules with polar bonds and a net dipole moment. The partial charges within nearby dipoles experience a mutual attraction and align in such a way that partial negative charges orient toward partial positive charges.

Time spent: 0 sec

Subject:
General Chemistry

QID: 401773

System:
Interactions of Chemical Substances

The primary structure of a protein is a series of amino acids that are connected in a long chain via C–N peptide bonds (Figure 1).

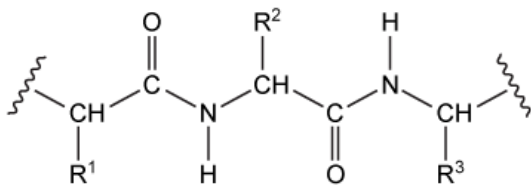


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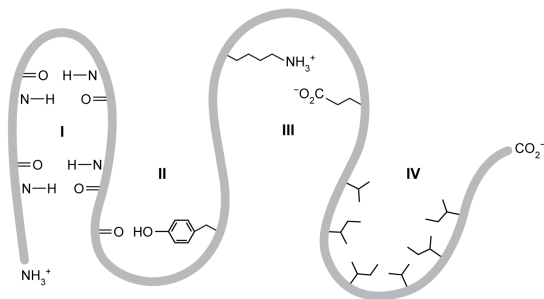


Figure 2 Intramolecular interactions between the peptide chain and amino acid side chains within a hypothetical protein

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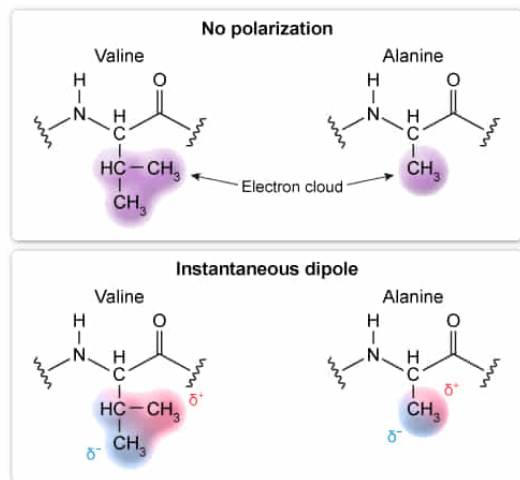
How will the cumulative strength of van der Waals interactions be affected in the hydrophobic region IV shown in Figure 2 if a protein variant is made that replaces valine with alanine in this region?

- ☒ A. The cumulative strength of van der Waals interactions will decrease because the side chain will have fewer hydrocarbon units. [66%]
- ☐ B. The cumulative strength of van der Waals interactions will increase because the side-chain of alanine has a permanent dipole moment. [3%]
- ☐ C. The cumulative strength of van der Waals interactions will increase because the region will have more hydrogen bonds. [2%]
- ☐ D. The cumulative strength of van der Waals interactions will remain the same because both valine and alanine are nonpolar. [27%]

Correct

66% answered correctly

Explanation:



- The larger the electron cloud, the more polarizable the molecule
 - The more polarizable the molecule, the stronger the van der Waals forces
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Van der Waals forces are noncovalent interactions between the dipoles of two neutral molecules. Van der Waals forces include:

- **Dipole-dipole interactions** (ie, attractions between two permanent dipoles)
- **Dipole-induced dipole interactions** (ie, attractions between a permanent dipole and an induced dipole)
- **London dispersion forces** (ie, attractions between two induced dipoles)

London dispersion forces (also called induced dipole-induced dipole interactions) are a weak mutual attraction resulting from the formation of **instantaneous dipoles** induced by momentary distortions in the "electron clouds" of molecules. As a result, London dispersion forces tend to be more pronounced in larger molecules because larger molecules have larger electron clouds that are more **polarizable**.

Because the side chains of valine and alanine are both nonpolar, lacking polar bonds with permanent dipoles, London dispersion forces are the only van der Waals interactions present in region IV of the protein shown in Figure 2. The side chain of valine is larger (ie, more hydrocarbon units) than the side chain of alanine. Consequently, replacing the larger valine with the smaller alanine in region IV will cause the extent of London dispersion forces (ie, van der Waals interactions) to decrease.

(Choice B) The side-chain of alanine is nonpolar and hydrophobic, and therefore does not have a permanent dipole moment.

(Choice C) Neither valine nor alanine have side chains with an -OH group or an -NH group to engage in **hydrogen bonding**. As such, the number of hydrogen bonds would not change with this amino acid replacement.

(Choice D) Although both valine and alanine are nonpolar, the extent of van der Waals interactions *will change* upon replacing valine with alanine because of the difference in the number of hydrocarbon units in the side chains.

Educational objective:

London dispersion forces are a type of van der Waals interaction by which molecules exhibit mutual attraction because of instantaneous dipoles induced by momentary distortions in the electron clouds around the molecules. Larger molecules have larger, more polarizable electron clouds, which results in stronger London dispersion forces.

Time spent:0 sec

Subject:
General Chemistry

QID:401774

System:
Interactions of Chemical Substances

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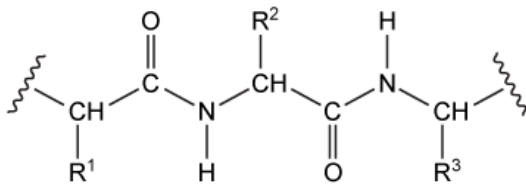


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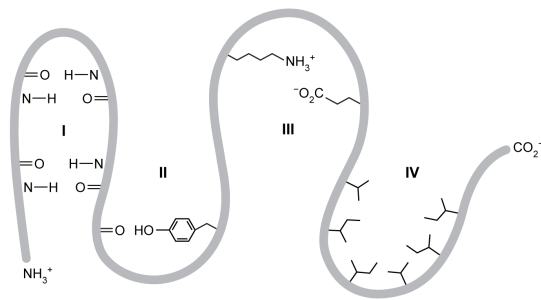


Figure 2 Intramolecular interactions between the peptide chain and amino acid side chains within a hypothetical protein

Intramolecular interactions play a key role in the folding of a protein into a three-dimensional structure. As such, the water solubility of a folded protein depends largely on the properties of the side chains exposed to the outer surface of the folded protein structure.

Suppose that one amino acid was substituted for a different amino acid on the surface of a folded protein. Which of the following substitutions would be least likely to affect the solubility of the folded protein in water? (Note: Assume that the amino acid substitutions do not alter the normal protein folding.)

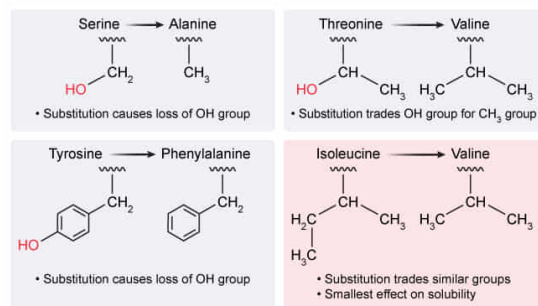
- ☐ A. Substitute alanine in place of serine [2%]
- ☐ B. Substitute valine in place of threonine [4%]
- ☐ C. Substitute phenylalanine in place of tyrosine [8%]
- ☒ D. Substitute valine in place of isoleucine [84%]

Correct

84% answered correctly

Explanation:

Side chain differences caused by amino acid substitutions



For a compound to be soluble in water, a sufficient number of attractive **intermolecular interactions** must be able to form between the structure of the compound and the structure of the water molecules to enable the compound molecules to become **solvated** (ie, individually surrounded by water molecules) and carried into solution. Because the O–H bonds in water are highly **polar** with a **strong dipole moment**, a compound will dissolve in water only if its structure has enough polar sites to form intermolecular attractions with the polar water molecules.

Hydrophobic ("water-fearing") molecules are sparingly soluble in water because they are predominantly **nonpolar** and cannot participate in **dipole-dipole interactions** or **hydrogen bonding** with water. Conversely, **hydrophilic** ("water-loving") molecules are soluble in water because they have enough polar bonds to attract the polar bonds in water molecules.

Accordingly, a protein is soluble in water if its surface can form enough **hydrophilic interactions** with the polar bonds in the water molecules. Structural changes that **increase polar sites** (or decrease nonpolar sites) on the protein surface **increase water solubility**, and vice versa. However, replacing one polar (or nonpolar) site with something that has similar characteristics causes little or no change in solubility.

Isoleucine and valine both have nonpolar side chains of similar size. Consequently, replacing isoleucine with valine would not significantly change the polarity of the protein surface, and this substitution would cause little (if any) change in the water solubility of the folded protein.

(Choices A, B, and C) Serine, threonine, and tyrosine all have side chains with an –OH group capable of hydrogen bonding. Replacing serine with alanine, threonine with valine, or tyrosine with phenylalanine would remove an –OH group, leaving a nonpolar side chain, decreasing the number of polar sites on the protein surface, and substantially decreasing the protein solubility.

Educational objective:

For a compound to be soluble in water, a sufficient number of attractive intermolecular interactions must be able to form between the structure of the compound and the structure of the water molecules. Hydrophilic molecules tend to be predominantly polar and soluble in water. Structural changes that decrease polar sites on a molecule decrease its water solubility.

Time spent: 0 sec

Subject:

General Chemistry

QID: 401775

System:

Interactions of Chemical Substances